CODE ANALYSIS — EXIT RECHECK (REV F) · P2-A-101 REV N / A-102 REV I · CURRENT DESIGN

Supersedes Rev E (drawn on A-101 Rev K: 20 ft tower, F1; bump-out at V2). Method and factors as Rev D / E (IBC 2021 T1004.5, 1005.3, 1005.5, 1030.2, T1017.2, T1006.2.1, 1007.1.1). Not a code review: architect + AHJ confirm (D-008).

1 WHAT CHANGED SINCE REV E (all ASSUMED unless cited)

- ST-2 = 30 ft drum (D-082): the 27.2 ft stair diagonal fits with 1.5 ft to spare, so Rev E finding F1 is CLOSED. Drum door 64 in at 235° (D-083); landing overlap left to the architect (D-085).
- East side = D-072 / plan Rev J (D-084): bump-out 42 ft south, X7 back on the wall, X11 new, 8 ft public corridor. Old NE stair strip = MECH (remote) (D-086).
- Bus loop cut (D-087); A-201 Rev L adds a colonnade + canopy in front of E1 (ASSUMED). Counts unchanged: E1 + X1-X11, 4 stairs x 76 in, 2,200 seats.

2 OCCUPANT LOAD — LEVEL 1 FIXED SPACES (Rev D method)

SPACE	SF (Rev N)	FACTOR	LOAD	
Boys + girls locker (2)	7,014	50	142	
Event lockers 1-4	3,600	50	72	
Flex / team assembly	1,767	15	118	
Flex / storage (SW)	972	300	4	
Equipment storage + room	1,000	300	4	
Mech / elec (7 rooms)	4,302	300	18	
First aid	120	150	1	
Concession	275	200	2	
Chair / table / stage storage (annex)	2,100	300	7	
Lower tier seats (telescopic)		seats	1100	
L1 without the event floor			**1,468**	**Rev E 1,467**
L2 worst case (D-052): seats, S&C, X-train, admin, loop, rail			**1,460**	**Rev E 1,460**

Change: mech rooms now 7 incl. the bay and MECH (remote) (+1 occupant). L2 unchanged.

3 EXIT CAPACITY (doors 0.15 in / occ., 64 in per pair)

FLOOR CASE	FLOOR	BUILDING	NEED in	E1 ≥ ½	RESULT
Sports (50 gross)	329	3,257	488.5	244.3	PASS
Tables + chairs (15 net)	1,095	4,023	603.4	301.7	PASS
Chairs only (7 net) DESIGN	**2,346**	**5,274**	**791.1**	**395.6**	**PASS**
Standing (5 net) margin	3,284	6,212	931.8	465.9	PASS

Provided: E1 512 in (8 pairs) + 11 exits x 64 = 1,216 in. Lose E1: others 704 in ≥ ½ of the worst need (1005.5). PASS.

4 LEVEL 2 STAIR CAPACITY (0.2 in / occ.)

- 1,460 x 0.2 = 292.0 in vs 4 x 76 = 304 in (spare 12.0). Lose one stair: 228 ≥ 146.0. Stair discharge doors 1,460 / 4 x 0.15 = 54.8 in each of 64. PASS. ST-2 now really holds the 76 in stair (D-082).

5 EXIT LOCATIONS (tested on the Rev N outline)

EXIT	POSITION	NOTE	OK
X1	W wall (0.0, 50.0)	team assembly (D-078)	PASS
X2	W wall (0.0, 98.0)	girls locker	PASS
X3	W wall (0.0, 182.0)	boys locker	PASS
X4	N wall (42.7, 252.0)	ST-1 discharge	PASS
X5	N wall (127.8, 252.0)	EXIT (N), bay clear	PASS
X6	drum E face (197.0, 262.5)	ST-2 30 ft drum, east face; clear of bay / annex	PASS
X7	E wall (210.0, 179.1)	EXIT (E) on the wall, north of the bump-out (D-084)	PASS
X8	E wall (210.0, 34.0)	east concourse	PASS
X9	S wall (183.7, 0.0)	ST-3 discharge (D-078)	PASS
X10	W wall (0.0, 32.0)	ST-4 discharge (D-078)	PASS
X11	E wall (210.0, 112.0)	public corridor (E), south of the bump-out (D-084)	PASS
E1	S wall bank (113.0, 0.0)	8-pair bank x 87.25-138.75	PASS

Test: door on the envelope line, whole 64 in leaf on a straight wall (not in an R 20 arc), clear of annex / bay / bump-out; X6 on the drum.

6 TRAVEL · COMMON PATH · SEPARATION

- L1 exit access (T1017.2, A sprinklered, 250 ft): worst point 111 ft direct to the nearest exit at (103, 141); routed path TBD. X7 path T1 133 ft. PASS.
- L2 (Rev E method): loop worst 160 ft incl. 15 ft seat access. PASS.
- Common path (T1006.2.1, 75 ft): women (2) 61 ft, men (2) 30 ft. Dead end 5.2 ft (20). Corridor 96 in (44). PASS.
- Stair separation (1007.1.1, 1/3 of the 328 ft diagonal = 109 ft): shortest 141 ft. PASS.

7 DISCHARGE (D-087: no bus loop)

- All exits discharge to open grade; no drive crosses a discharge. Public-way route and grading: parcel TBD (D-006).
- E1 passes the colonnade: 4 gaps x 13 ft = 624 in ≥ 512 in; then the 40 ft walk around / through the portal (D-066). PASS.

8 FINDINGS (options; not redesigned)

- F1 Colonnade column on the E1 / portal axis (x 113), 10 ft out. Width passes. Options: (a) keep; (b) even column count so the axis is open; (c) wider centre bay. Shane / architect.
- F2 X6 discharges between the drum and the bay corner (3 ft apart, west of the door). Confirm the NE walk / grading so X6 reaches open ground (D-006).
- F3 Mechanical 4.95 % vs 5 % (D-072, MEP OPEN). Not an egress item; listed for the set.

9 OPEN FOR THE ARCHITECT / AHJ

- Everything in Rev D section 12 stands: editions + AHJ (D-008), construction type, smoke-protected seating, tier structure, door widths. Drum door landing (D-085). Curved-shell structure.

Sources: p2_egress_rev_n.py; p2_plan_rev_n_geom.py (A-101 Rev N / A-102 Rev I); phase2_facade_srm.yaml; P2-G-002 Rev D / E; IBC 2021 as cited; D-052, D-054, D-066, D-072, D-082..D-087.

PRELIMINARY — CONCEPTUAL DESIGN — NOT FOR CONSTRUCTION

PROJECT	PHASE	SHEET TITLE	SCALE	DATE	REV	DRAWN BY	SHEET
Hazel Green Regional Athletic Complex Trojan Horse Arena · Hazel Green area, Madison County, Alabama (site TBD, D-006)	**PHASE 2**	**CODE ANALYSIS EXIT RECHECK**	**NTS**	**October 2026**	**F**	Drawn by KEYSTONE (AI) for Shane Brazelton	**P2-G-002**